Q3.a

$$X(K)=\sum_{n=0}^{N-1} x(n)\, e^{-j\frac{2\pi nk}{N}}$$

$$X(k+N)=\sum x(n)\, e^{-j\frac{2\pi n}{N}(k+N)}$$

$$\sum x(n)\, e^{-j\frac{2\pi nk}{N}} \cdot e^{j\frac{2\pi nN}{N}}$$

Modern Academy

for Engineering and Technology in Maadi
Electronic Engineering and
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QUESTIONS FOR THE FINAL WRITTEN EXAM

Subject: (ELCN425 & ELCn425) Digital Signal Processing
Examiners: Dr. Samir M. Kamal and Dr. Sara M. Hassan

Spec.: Communications & Computer
Time: 2 Hours

Number of Pages: One | Number of Questions: 4 | Answer All Questions

$y(n)=x(n)+0.5x(n-1)+1.5y(n-1)+0.5y(n-3)$

Question 1 **(10 points)**

a- For LTIS define (using equations, figures and examples): Difference equation and Transfer function.

b- For the following transfer function: $H(z)=\frac{z^2(z+0.5)}{(z-0.5)(z-1)^2}$ $(z-0.5)(z^2+2z+1)$

(i) Plot the pole - zero diagram of H (z).

(ii) Find the difference equation of this filter, y (n). $z^3+2z^2+z-0.5z^2-z-0.5$

(iii) Represent the filtering operation in a block diagram form. $z^3+1.5z^2-0.5$

(iv) Using the partial fration method find "4" values for the impulse response of this system. (Use the given table) $h(n)=2(0.5)^n+3n+17$ $0\to 19$ $2\to 23.5$ $1\to 21$ $3\to 26.25$

Question 2 **(10 points)**

a- Define (using equations and/or figures) the following items: Correlation operation and Convolution operation.

b- Given two sets of samples $x_1(n)=\{2, 2, 0, 1\}$ and $x_2(n)=\{1, 2, 0, 1\}$.

(i) Find the linear cross- correlation between $x_1(n)$ and $x_2(n)$. Shift Left & Rotate

(ii) Plot the obtained linear correlation v.s. the lag number, j. $x_2(-n)=\{1,1,0,2\}$ cyclic

(iii) Find their convolution, y(n). $x_1(n)=\{2,2,0,1\}$

(iv) Plot the obtained convolution v.s. n.

$y(n)=(6,5,3,3)$

$R_{12}(j)$: 1, 0.571, 0.285, 0.285, 0.428, 0.285, 0.285

(10 points)

Question 3

(i) Show that the DFT, X(K), of a sequence x(n) is periodic.

(ii) For the following sequence: $x(n)=\{1, 0, 1, 1\}$:

(i) Using the direct method, FFT, find the frequency components of this sequence. $X(K)=\{3, j, 1, -j\}$

(ii) Check your results obtained in part (i) by calculating x(n) using the IDFT method. $\{1, 0, 1, 1\}$

(iii) Using your results, verify Parseval's theorem. $3=12/4$

(iv) What is the computational saving if we use FFT algorithm instead of DFT algorithm? DFT: add 12, mul 16; FFT: 8, 4 (10 points); Save 4, 12

Question 4

a- Derive the sampling condition. $F_S \geq 2f_{max}$

b- For the signal $x(t)=10\cos(1000\pi t)$: $f=500$ Hz

(i) If it is sampled with Fs = 2000 sample/sec, is it possible to reconstruct this signal back from its samples successfully? Construct $\to$ 1000 Hz $F_N=\frac{F}{F_S}=\frac{1}{4}=\frac{F}{NF}=0.25$

(ii) Find the Nyquist frequency, F_N, for this signal.

(iii) Using the obtained Nyquist frequency, F_N find an expression for the corresponding sequence, $x(n)=10\cos(\frac{\pi n}{2})$

(iv) Find the values of x(n) in the first cycle.

Z-transforms of some required sequences.

Discrete-time sequence x(n), n ≥0	z-transform X(z)	Note
kn	$\frac{kz}{(z-1)^2}$	k and α are constants
k	$\frac{kz}{z-1}$	
$k\alpha^n$	$\frac{kz}{z-\alpha}$	

Good Luck